

13.0 FABRICATION OF REINFORCEMENT

- 13.01 SPLICING OF REINFORCEMENT BARS IS PERMITTED:-
- AT LOCATIONS SPECIFIED/SHOWN IN DRAWINGS.
 - AT OTHER LOCATIONS, IF LENGTH AS PER DRAWINGS IS MORE THAN THE MAXIMUM AVAILABLE LENGTH OF SINGLE UN-SPLICED BAR, SUBJECT TO APPROVAL OF SUCH LOCATIONS.
- 13.02 UNLESS SPECIFIED ON DRAWING OR OTHERWISE PERMITTED, ALL SPLICES SHALL BE LAPPED SPLICES WITHOUT WELDING.
- 13.03 LENGTH OF LAPS FOR LAPPED SPLICES (WITHOUT WELDING) SHALL BE AS PER TABLE-1.
- 13.04 WHERE DETAILS AND SPECIFIC LOCATIONS OF SPLICES ARE NOT SHOWN ON DRAWINGS, LAP SPLICES SHALL BE SO PROVIDED THAT-
- NOT MORE THAN 50% OF THE TOTAL BARS AT ANY FACE OF THE MEMBER ARE SPLICED AT ANY SECTION.
 - CLEAR SPACE FROM END OF ONE SPlice TO THE BEGINNING OF NEXT SPlice OF ANY BAR SHALL NOT BE LESS THAN 0.3 TIMES THE DEVELOPMENT LENGTH OF THE BAR.
- 13.05 NO SPLICES SHALL HAVE LAP LENGTH LESS THAN 300mm.
- 13.06 NO LAP SPLICES SHALL BE PERMITTED IN A HANGER.
- 13.07 NO REINFORCING BAR SHALL BE KINKED OR CRANKED EXCEPT AS SHOWN ON DRAWINGS OR APPROVED OTHERWISE.
- 13.08 WHERE, FOR ANY REASON, IT IS NOT POSSIBLE TO CONFIRM TO THE REQUIREMENTS OF 12.04 ABOVE, LENGTH OF LAPPED SPlice AS SPECIFIED SHALL BE INCREASED BY 30%
- 13.09 LAPS FOR REINFORCEMENT IN COMPRESSION IN BEAMS SHALL BE ACCOUNTED ONLY AS STRAIGHT LENGTHS AND NO HOOKS OR BENDS ARE ACCOUNTED FOR COMPRESSION LAPS OR ANCHORAGES.
- 13.10 STANDARD HOOKS OR BENDS SHALL BE PROVIDED WHERE NECESSARY IN TENSION REINFORCEMENTS IN BEAMS, COLUMNS, SLABS, ETC.
- * 13.11 WHEN DIAMETER OF THE BARS IS 20MM OR HIGHER, MECHANICAL COUPLERS AS PER IS 16172 SHALL BE USED TO EXTEND BARS. COUPLER CRITERIA AND TEST REPTS SHALL BE PROVIDED. IF LAPPING OF BARS IS REQUIRED IN EXCEPTIONAL CASE, RELEVANT CLAUSES OF IS 13920 SHALL APPLY.

14.00 POSITIONING OF REINFORCEMENT

- 14.01 ALL REINFORCING BARS SHALL BE PLACED IN FORMWORK OF CONCRETE WITH SPECIFIED CLEAR COVER AS PER TABLE 2 OF THIS DRAWING.
- 14.02 WHERE BAR SPACING HAS BEEN INDICATED IN A DEFINED RANGE WITHOUT SPECIFYING NUMBER OF SUCH BARS, THE BARS SHALL BE SO ARRANGED THAT THE TOTAL NUMBER OF BARS SHALL BE EQUAL TO THE RANGE (OR SPREAD) DIVIDED BY SPACING, PLUS ONE, ROUNDED TO THE NEAREST WHOLE. EXAMPLE:-



NO. OF BARS = ((5000/150) + 1) = 33.33 + 1 = 34

- 14.03 WHERE BAR SPACING HAS NOT BEEN INDICATED, THE TOTAL NUMBER OF BARS (IN A SPECIFIED LAYER AS SHOWN IN DETAILS) MAY BE EQUALLY SPACED, PROVIDED SUCH ARRANGEMENT DOES NOT HINDER EASY PLACING AND COMPACTION OF CONCRETE.
- 14.04 UNLESS OTHERWISE SPECIFIED, SPACER BARS SHALL BE PROVIDED TO MAINTAIN CONSTANT SPACING BETWEEN CONSECUTIVE LAYERS OF REINFORCEMENT. SPACER BARS SHALL BE 20mm DIA. OR SAME DIA. AS LARGEST DIAMETER OF BAR USED, WHICHEVER IS GREATER.

- *14.05 WHERE SPECIFIED LOCATION OF REINFORCING BARS IS SUCH THAT IT CAN NOT BE MAINTAINED EFFECTIVELY USING COVER BLOCKS OR SPACER BARS, CHAIRS SHALL BE PROVIDED AS APPROPRIATE.

15.00 PLACING AND COMPACTION OF CONCRETE

- 15.01 CONCRETE IN ALL ELEMENTS SHALL BE PLACED IN SINGLE POUR BETWEEN SPECIFIED/PERMITTED CONSTRUCTION JOINTS.
- 15.02 UNLESS OTHERWISE SPECIFIED AND DETAILED IN DRAWINGS, NO HORIZONTAL CONSTRUCTION JOINTS SHALL BE PERMITTED IN SLABS AND BEAMS.
- 15.03 THE FOLLOWING RESTRICTION APPLIES WHERE CONCRETE OF DIFFERENT GRADES ARE INVOLVED -
- THE CONCRETE GRADE IN BEAMS AND SLABS MEETING AT COLUMN SHALL BE THE SAME AS SPECIFIED FOR COLUMN WITHIN AN AREA AROUND THE COLUMN EQUAL TO 4 TIMES THE CROSS - SECTIONAL AREA OF THE COLUMN. SUBJECT TO THE PROVISION THAT THIS GRADE OF CONCRETE SHALL EXTEND ATLEAST FOR A DISTANCE EQUAL TO THE DEPTH OF BEAM OR SLAB FROM THE FACE OF COLUMN, FOR EACH BEAM OR SLAB MEETING AT THE COLUMN.

16.0 EMBEDMENTS, INSERTS AND FITTINGS

- 16.01 NO CUT-OUTS, POCKETS, CHASES ETC., SHALL BE MADE IN HARDENED CONCRETE WITHOUT APPROVAL. ALL INSERTS, HOLDFASTS ETC., REQUIRED IN THE CONCRETE SHALL BE PROVIDED DURING CONSTRUCTION.
- 16.02 CONDUITS, PIPES, SLEEVES AND OTHER INSERTS ARE NOT PERMITTED TO BE EMBEDDED IN CONCRETE UNLESS SHOWN IN THE STRUCTURAL DRAWINGS OR OTHERWISE SPECIFICALLY APPROVED.
- 16.03 NOTWITHSTANDING ANY DETAILS SHOWN IN SERVICES DRGS, NO CONDUITS OR PIPES SHALL BE LAID TO BE EMBEDDED IN CONCRETE ALONG A MAIN REINFORCING BAR WITHOUT ADEQUATE CONCRETE COVER/SPACE AROUND THEM.
- 16.04 NO CONDUITS, PIPES, SLEEVES AND OTHER INSERTS SHALL BE PARTIALLY EMBEDDED IN CONCRETE UNLESS SO INDICATED IN STRUCTURAL DRAWINGS.
- 16.05 NO CONDUITS, PIPES, SLEEVES OR OTHER INSERTS AND EMBEDMENTS IN CONCRETE SHALL BE OF MATERIALS THAT ARE POTENTIALLY OF DELETERIOUS EFFECT ON CONCRETE OR ON REINFORCING STEEL.

- 16.06 THE CLEAR SPACING OF EMBEDDED CONDUITS, PIPES, SLEEVES, ETC., SHALL NOT BE LESS THAN TWICE THEIR DIA. AND CLEAR CONCRETE COVER NOT LESS THAN 20mm IF UNEXPOSED TO WEATHER AND 40mm IF EXPOSED TO WEATHER OR TO GROUND.
- 16.07 NONE OF THE EMBEDMENTS, INSERTS OR FITTINGS SHALL BE WELDED TO REINF. BARS UNLESS SPECIFICALLY APPROVED.
- 16.08 NONE OF THE ABOVE TYPE OF EMBEDMENTS SHALL BE PERMITTED IN COLS. UNLESS SO SHOWN IN STRUCTURAL DRGS.
- 16.09 NO CONDUITS, PIPES, OR SLEEVES OF LARGER DIA THAN ONE-THIRD THE THICKNESS OF SLAB OR BEAM SHALL BE PERMITTED IN THE LATTER.
- 16.10 NOTWITHSTANDING REQUIREMENTS SPECIFIED IN RELEVANT SERVICES DRGS, NO EMBEDMENTS INSERTS OR FITTINGS SHALL BE PLACED IN STRUCTURAL CONCRETE IN A MANNER WHICH MAY ADVERSELY AFFECT THE STRENGTH OF THE MEMBER.

17.00 SEQUENCE OF PLANNING AND TECHNIQUES

- 17.01 ALL TEMPORARY AND PERMANENT WORKS REQUIRED FOR THE COMPLETION OF THE PROJECT SHALL BE DONE IN ACCORDANCE WITH SPECIFICATIONS, DRAWINGS, INSTRUCTIONS AND GOOD ENGINEERING NORMS, AND WITHIN THE SPECIFIED PERIOD.
- 17.02 ADEQUACY OF NECESSARY MEASURES AND SAFETY OF ALL SUCH WORKS, PROPERTY AND PERSONNEL INVOLVED SHALL BE ENSURED THROUGHOUT THE TENURE OF EXECUTION OF THE WORKS.
- 17.03 THE SEQUENCE, PLANNING AND TECHNIQUES OF CONSTRUCTION ADOPTED SHALL BE SUCH AS TO ENSURE THAT-
- AT NO STAGE ANY PART OF STRUCTURES AND THEIR COMPONENT ELEMENTS ARE STRESSED BEYOND LIMITS OF THE STRENGTH OF MEMBERS CONCERNED.
 - NON-STRUCTURAL ELEMENTS SUCH AS PARTITIONS AND FINISHES ARE NOT SUBJECTED TO DISTRESS/CRACKING/ DEFORMATION DUE TO SHORT OR LONG TERM STRUCTURAL DEFORMATIONS.
 - AT NO STAGE ANY PART OF STRUCTURE OF A STRUCTURAL MEMBER OR A NON-STRUCTURAL ELEMENT OF THE BUILDING REMAINS IN A CONDITION OF INSTABILITY OR POTENTIAL INSTABILITY.
- 17.04 WHERE A PART OF STRUCTURE JOINS ANOTHER THROUGH EXPANSION JOINT, CONSTRUCTION JOINT, ETC., THE SEQUENCE AND TECHNIQUE OF CONSTRUCTION SHALL TAKE INTO ACCOUNT CO-ORDINATION NECESSARY BETWEEN THE WORKS IF DIFFERENT CONTRACTORS/AGENCIES ARE INVOLVED.
- 17.05 FOR LONG-SPAN ELEMENTS AND OTHER MEMBERS WHERE ANTICIPATED SHORT AND LONG TERM DEFLECTIONS ARE LARGE, CAMBER SHALL BE PROVIDED IN THE FORM WORK IN CONSULTATION WITH THE STRUCTURAL CONSULTANTS.
- 17.06 IN CASE OF HANGERS, THE STRUCTURAL MEMBERS AT LOWER END OF HANGERS SHALL BE KEPT EFFECTIVELY SUPPORTED FROM GROUND OR ANY OTHER FIRM BASE UNTIL THE STRUCTURAL MEMBERS AT THE UPPER END HAVE BEEN CONSTRUCTED AND READY.

18.00 WATER PROOFING

- 18.01 WATER PROOFING SHALL BE DONE FOR: FOUNDATION, STP, SUMP TANKS, OVERHEAD TANK, BASEMENT WALLS, BASEMENT, TOILET SLABS, TERRACE, AS PER QST SPECIFICATIONS TO ENSURE WATER SHALL NOT INGRESS INTO STRUCTURAL ELEMENT.

19.00 CURING

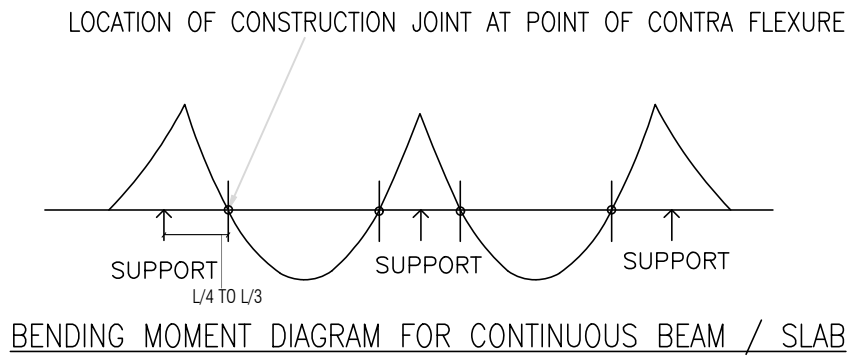
- 19.01 EXPOSED SURFACES OF CONCRETE SHALL BE KEPT CONTINUOUSLY IN DAMP OR WET CONDITION BY PONDING OR BY COVERING WITH A LAYER OF ANY MATERIAL SUITABLE.
- 19.02 CURING COMPOUND MAY BE USED MINIMUM CURING PERIOD FOR CONCRETE WITH ORDINARY PORTLAND CEMENT - 7 DAYS WITH BLENDED CEMENT - 14 DAYS
- 19.03 CURING FOR HIGHER GRADE OF CONCRETE(M35 UPWARDS) TO START AFTER 4 Hrs. OF LAYING CONCRETE.
- 19.04 THE PERIOD OF CURING SHALL NOT BE LESS THAN 10 DAYS FOR CONCRETE EXPOSED TO DRY & HOT WEATHER CONDITIONS.

20.00 DESHUTTERING TIME

- 20.01 TYPE OF FORM WORK FOR : MINIMUM PERIOD BEFORE STRIKING (EXCLUDING THE DAY OF CASTING)
- SLABS UPTO 4.5M 7 DAYS OVER 4.5M 14 DAYS
 - BEAMS UPTO 6.0M 14 DAYS OVER 6.0M 21 DAYS
 - WALL/BEAM/COLUMN VERTICAL FACES 24 HOURS
- 20.02 SHUTTERING CAN BE RELEASED EARLIER THAN SPECIFIED ABOVE PROVIDED THE PROPS ARE RETAINED AT REASONABLE SPACINGS AND CONCRETE ATTAINS THE FULL STRENGTH AT THE TIME OF DESHUTTERING AS PER IS 456 - 2000. DESHUTTERING SHALL BE DONE BY PROVIDING DAMPENING MATERIAL BELOW IN ORDER TO REDUCE THE IMPACT AT THE LOWER LEVEL.
- 20.03 SIMPLY SUPPORTED AND CONTINUOUS SPANS SHALL BE DESHUTTERED FROM MID SPAN TOWARDS SUPPORTS. CANTILEVERS SHALL BE DESHUTTERED FROM FREE END TOWARDS SUPPORTS.
- 20.04 SLAB/BEAMS SHALL BE REPROPPED UNTIL THE NEXT LEVEL SLAB IS CAST AND DESHUTTERED.

21.00 CONSTRUCTION JOINTS

- 21.01 SLABS : AT ONE THIRD OF THE SPAN FROM SUPPORT.
BEAMS : AT ONE THIRD OF THE SPAN FROM SUPPORT.
- 21.02 IF ANY CROSS BEAMS ARE LOCATED IN THIS REGION THE JOINT SHALL BE LOCATED TWO TIMES THE WIDTH OF THE BEAM AWAY FROM THE BEAM.
- 21.03 RETAINING WALL : AT THE JUNCTION OF THE WALL AND FOOTING WITH A WALL KEY JOINT AND AT MIDDLE OF VERTICAL SPAN, JOINTS TO BE AVOIDED.
- 21.04 RAFT SLAB : AT ONE THIRD OF THE SPAN FROM SUPPORT.
CONSTRUCTION JOINTS SHALL BE STRAIGHT.
- 21.05 CANTILEVERS : CONSTRUCTION JOINT SHALL NOT BE PROVIDED IN CANTILEVERS.
WATER STOPPER & NECESSARY MEASURES SHALL BE TAKEN CARE ALL CONSTRUCTION JOINTS AS PER RELEVANT IS CODES.



22.00 ADMIXTURE

- 22.01 CONCRETE ADMIXTURE SHALL BE USED TO IMPROVE WORKABILITY OR STRENGTH AS MANUFACTURER SPECIFICATION.

- 22.02 WATERPROOFING ADMIXTURE SHALL BE USED FOR WATER RETAINING STRUCTURES, PODIUM, TERRACES, AND LIQUID STORAGE STRUCTURES.

23.00 GROUT

- 23.01 CEMENT BASED NON-SHRINK GROUT WITH A MINIMUM 7 - DAY CUBE STRENGTH OF 35 N/MM² SHALL BE USED.

24.00 ANCHORE BOLTS

- 24.01 ANCHORE BOLTS GRADE 8.8 MINIMUM, YIELD STRENGTH 640 N/MM² REQUIRED. SPECIAL INSPECTION REQUIRED AND SET ALL ANCORE BOLTS BY TEMPLATE.

25.00 MISCELLANEOUS

- 25.01 REFER TO ARCHITECTURAL DRAWINGS FOR WALL OPENINGS, ARCHITECTURAL FEATURES AND DIMENSIONS NOT SHOWN.REFER TO MEP DRWINGS FOR SIZE AND LOCATION OF DUCT OPENINGS, PIPING, CONDUITS, ETC.
- 25.02 NOT SHOWN.REFER TO SEPARATE SPECIFICATIONS FOR INFORMATIONS NOT CONTAINED IN THESE GENERAL NOTES.

26.00 LIQUID RETAINING STRUCTURES

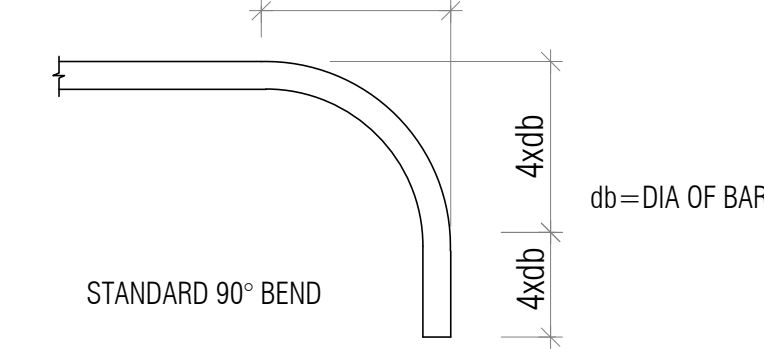
- 26.01 CONCRETE SHALL BE OF GRADE M30 WITH MINIMUM CEMENT CONTENT OF 330 kg/Cum.M AS PER IS : 3370.PART 1 TANKS SHALL BE TESTED FOR WATER TIGHTNESS AS PER IS : 3370 PART-1.
- 26.02 ALL STRUCTURAL STEEL EMBEDMENTS , LADDERS ETC. , SHALL BE PROVIDED WITH 2 COATS ANTICORROSIVE PAINT OVER 2 COATS OF PRIMER.
- 26.03 BACKFILLING AROUND THE WATER TANK SHALL BE DONE AFTER THE TESTING OF THE TANK IS COMPLETE WITH THE APPROVAL OF THE ENGINEER - IN - CHARGE.
- 26.04 CARE SHALL BE TAKEN AT ALL TIMES TO ENSURE THAT WATER BARS ARE NOT PERFORATED OR DAMAGED IN ANY WAY AND CONCRETE SHALL BE CAREFULLY PLACED AND COMPACTED AROUND THE WATER BAR TO ENSURE DENSE VOID-FREE IMPERVIOUS CONCRETE.
- 26.05 ANY BOLT OR FASTENING EMBEDDED IN OR PASSING THROUGH CONCRETE SHALL BE TO THE APPROVAL OF THE ENGINEER - IN - CHARGE AND SHALL NOT IMPAIR THE WATERTIGHTNESS OF THE STRUCTURE. ALL PIPE SLEEVES SHALL BE PROVIDED WITH PUDDLE FLANGES. THE CUTTING OR DRILLING OF HOLES IN WATER TANKS IS STRICTLY PROHIBITED. ALL THE ABOVE DETAILS REFER QST STANDARD DRAWINGS. FOR MORE CLARITY.

27.00 BRICKWORK/ BLOCKWORK.

- 27.01 ALL BRICKWORK/BLOCKWORK WALLS SHALL CONFIRM TO THE SPECIFICATION REFER QST MANNUALS FOR DETAILS.

28.00 DEVELOPMENT AND ANCHORAGE LENGTHS

- 28.01 'Ld' FOR REINFORCEMENT IN COMPRESSION SHALL BE ACCOUNTED ONLY AS STRAIGHT LENGTHS AND NO HOOKS OR BENDS ARE ACCOUNTED FOR COMPRESSION LAPS OR ANCHORAGES.
- 28.02 STANDARD HOOKS OR BENDS SHALL BE PROVIDED WHERE NECESSARY IN TENSION REINFORCEMENTS IN BEAMS, COLUMNS, SLABS, ETC.



- 28.03 THE FOLLOWING DEVELOPMENT LENGTHS ARE TO BE PROVIDED FOR REINFORCEMENT IN TENSION AND COMPRESSION FOR M20, M25, M30, M35 & M40 GRADE OF CONCRETE AND Fe415, Fe500/500D & Fe550 GRADE STEEL.

TABLE 1

DIA. IN mm	LD FOR FY-415				
	M20 GRADE	M25 GRADE	M30 GRADE	M35 GRADE	M40 & ABOVE GRADE
8	400	350	300	275	250
10	500	425	380	350	300
12	565	485	455	400	355
16	760	645	610	535	475
20	940	810	755	665	600
25	1175	1010	940	830	750
32	1510	1290	1210	1065	950

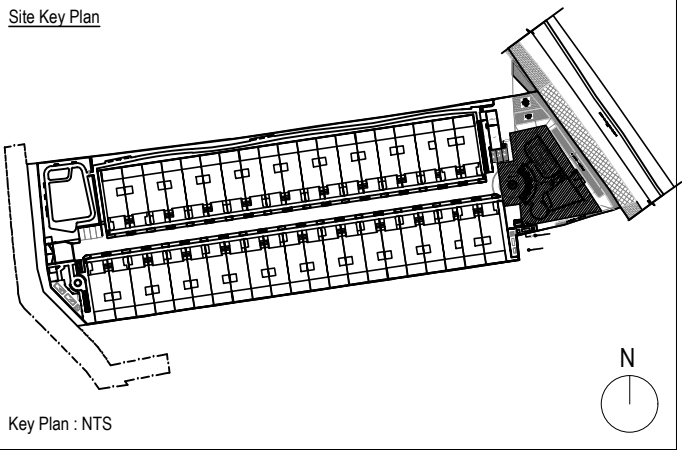
DIA. IN mm	LD FOR FY-500				
	M20 GRADE	M25 GRADE	M30 GRADE	M35 GRADE	M40 GRADE9.
8	455	385	365	325	285
10	565	485	455	400	360
12	680	585	545	480	430
16	910	780	725	650	575
20	1135	975	910	800	715
25	1415	1215	1135	1000	900
32	1815	1555	1450	1280	1150

DIA. IN mm	LD FOR FY-550				
	M20 GRADE	M25 GRADE	M30 GRADE	M35 GRADE	M40 GRADE
8	500	425	400	350	325
10	625	550	500	450	400
12	750	650	600	525	475
16	1000	850	800	700	625
20	1250	1050	1000	875	800
25	1550	1350	1250	1100	1000
32	2000	1700	1600	1400	1250

TABLE 2

SL NO	MEMBER	CLEAR COVER	CONCRETE MIX	GRADE OF STEEL
1.	FOOTING	50	M30	Fe550
2.	COLUMN PEDESTAL	40	M30	Fe550
3.	COLUMNS/WALL ABOVE PLINTH	40	M30	Fe550
4.	PLINTH BEAM	30	M30	Fe550
5.	RETAINING WALLS (IF ANY)	30	M30	Fe550
6.	COLUMNS/WALL BELOW PLINTH (* MINIMUM COVER TO LONGITUDINAL BAR & NOT TO TIES)	40 *	M30	Fe550
7.	BEAM IN SUPERSTRUCTURE	30	M30	Fe550
8.	SLAB IN SUPERSTRUCTURE	30	M30	Fe550
9.	LINTELS	25	M30	Fe550
10.	OVERHEAD WATER TANK BOTTOM SLAB	30	M30	Fe550
11.	OVERHEAD WATER TANK WALLS	30	M30	Fe550
12.	OVERHEAD WATER TANK TOP SLAB	30	M30	Fe550
13.	UNDER GROUND WATER TANK BOTTOM SLAB	50	M30	Fe550
14.	UNDER GROUND WATER TANK WALLS	40	M30	Fe550
15.	UNDER GROUND WATER TANK TOP SLAB	30	M30	Fe550
16.	STAIR CASE SLABS	30	M30	Fe550

IN ALL OTHER MEMBERS, UNLESS OTHERWISE SPECIFIED, CLEAR COVER TO OUTERMOST LAYER OF REINFORCEMENT SHALL BE 25MM OR 1-1/2 TIMES THE DIAMETER OF BAR WHICHEVER IS GREATER.



- General Notes
- ALL DIMENSIONS AND LEVELS ON THE BUILDING SITE ARE TO BE CHECKED BEFORE COMMENCING WITH ANY WORK.
 - ANY DISCREPANCY IN THE DRAWINGS, DIMENSIONS & SPECIFICATIONS TO BE BROUGHT TO THE ATTENTION OF THE ARCHITECTS AND RELEVANT PARTY PRIOR TO ANY WORK BEING STARTED.
 - ONLY FIGURED DIMENSIONS TO BE FOLLOWED - SCALED DIMENSIONS OF DRAWINGS WILL NOT BE PERMITTED.

- Key / Additional Notes
- READ THIS DRG. ALONG WITH GENERAL NOTES DWG. NO. SE-10(SHEET-1) & SE-100(SHEET-2)

Rev	Description	Date	Issued by
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